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Inventor(s)	Allmandinger; Alex et al.

Aerodynamic device control

Abstract

An aerodynamics control system for a vehicle may include a repositionable aerodynamic device disposed at a portion of the vehicle, a controller operably coupled to components or sensors of the vehicle to receive information including vehicle performance data and position information for the aerodynamic device, and a vehicle location sensor determining location information for the vehicle. The controller stores the vehicle performance data and the position information in association with the location information for each of a plurality of locations. Responsive to detecting an approach of the vehicle to one of the locations, the controller provides a control instruction to position the aerodynamic device based on recorded vehicle performance data and recorded position information associated with the one of the locations.

Inventors: Allmandinger; Alex (Ferndale, MI), Walsh; Robert (Novi, MI), Bastanipour; Michael (Ferndale, MI), Pakkam; Sriram (Royal Oak, MI)

Applicant: Ford Global Technologies, LLC (Dearborn, MI)

Family ID: 83998234

Assignee: Ford Global Technologies, LLC (Dearborn, MI)

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Primary Examiner: Lee; James J

Assistant Examiner: Huber; Melanie G

Attorney, Agent or Firm: Burr & Forman

Background/Summary

TECHNICAL FIELD

(1) Example embodiments generally relate to vehicle aerodynamics and, more particularly, relate to controllable aerodynamic devices for which the control is based on situational awareness.

BACKGROUND

(2) Vehicles, and especially high performance vehicles, often employ aerodynamic devices that are designed to improve aerodynamic performance. These aerodynamic devices are typically static. The static positioning of aerodynamic devices provides increased downforce (decreasing lift)

during turns, which generally improves handling and stability. Additionally, setting aerodynamic balance (i.e., the location of the center of pressures or centroid of the aerodynamic force) is also generally limited to being adjusted before driving. However, the optimal aerodynamic balance is different for different cornering radii, vehicle weight, and road conditions.

(3) Because aerodynamic devices are typically static, they may provide improved performance over only a limited set of vehicle conditions. As such, these static aerodynamic devices may be designed to be set at positions that provide increases in average performance. Different positions may be employed, each statically positioned, and vehicle performance may be compared to provide the best average position to optimize vehicle performance.

(4) Thus, it may be desirable to develop aerodynamic devices that addresses some of the shortcomings discussed above.

BRIEF SUMMARY OF SOME EXAMPLES

(5) In accordance with an example embodiment, an aerodynamics control system for a vehicle may be provided. The aerodynamics control system may include a repositionable aerodynamic device disposed at a portion of the vehicle, a controller operably coupled to components or sensors of the vehicle to receive information including vehicle performance data and position information for the aerodynamic device, and a vehicle location sensor determining location information for the vehicle. The controller stores the vehicle performance data and the position information in association with the location information for each of a plurality of locations. Responsive to detecting an approach of the vehicle to one of the locations, the controller provides a control instruction to position the aerodynamic device based on recorded vehicle performance data and recorded position information associated with the one of the locations.

(6) In another example embodiment, a method of providing vehicle stability control to a vehicle having an aerodynamic device disposed on the vehicle is provided. The method may include receiving status information and location information from components or a sensor network of the vehicle, determining whether the location information is a known location, recording the location information and status information as path data for a path segment corresponding to the location information, and classifying the path segment in response to the location information not being known. The method may also include controlling the aerodynamic device based on a selected control algorithm in response to the location information being known. The selected control algorithm may be selected based on a classification of the path segment.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. 1 illustrates a block diagram of a vehicle control system in accordance with an example embodiment;

(3) FIG. 2 illustrates a block diagram of some components of an aerodynamic control system of the vehicle control system of FIG. 1 in accordance with an example embodiment;

(4) FIG. 3 illustrates a block diagram of how a controller may select control algorithms in accordance with an example embodiment;

(5) FIG. 4 illustrates a block diagram of a method of providing vehicle stability control to a vehicle having an aerodynamic device assembly disposed on the vehicle in accordance with an example embodiment; and

(6) FIG. 5 illustrates sub-segments and phases applicable to each sub-segment in accordance with an example embodiment.

DETAILED DESCRIPTION

(7) Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.

(8) Some example embodiments described herein may address the problems described above. In this regard, for example, some example embodiments may provide an improved system for aerodynamic control that includes an aerodynamic control module capable of being modified while driving to optimize drag, lift and other aerodynamic properties across various specific vehicle conditions. In some cases, the aerodynamic control module may utilize location information and other sensor information to appreciate current and/or future location so that active aerodynamic device positioning can be accomplished while driving. Moreover, in some cases, example embodiments may enable tuning (e.g., via machine learning) that occurs based on recognition of similar or the same conditions as those previously encountered, so that active changes may be made to optimize performance. Optimizing aerodynamic device positions for instantaneously determined conditions and situations can yield benefits in both customer confidence and vehicle capability with less compromise. As a result, vehicle performance and driver satisfaction may also be improved.

(9) FIG. 1 illustrates a block diagram of an aerodynamic control system **100** of an example embodiment. The components of the aerodynamic control system **100** may be incorporated into a vehicle **110** (e.g., via being operably coupled to a chassis of the vehicle **110** and/or electronic control systems of the vehicle **110**). Of note, although the components of FIG. 1 may be connected to the chassis of the vehicle **110**, it should be appreciated that such connection may be either direct or indirect. Moreover, some of the components of the aerodynamic control system **100** may be connected to the vehicle **110** via intermediate connections to other components either of the chassis or of other systems or components.

(10) The aerodynamic control system **100** may include an aerodynamic device assembly **120** that may include one or more aerodynamic devices (e.g., a wing, strake, vent, airfoil, air guide, or the like) that are movable responsive to control initiated by the aerodynamic control system **100** while the vehicle **110** is in motion. In its simplest form, the aerodynamic device assembly **120** may include a single aerodynamic device (e.g., a rear wing) that can be repositioned to change the angle of attack of the device while driving, or to change the device between a deployed state and a retracted state. Changes between the deployed state and retracted state, as well as changes to the angle of attack, may all be considered to be state changes for the aerodynamic device assembly **120**. The aerodynamic control system **100** may include programming and programmatically controlled physical structures that enable the initiation of changes to the state of the aerodynamic device assembly **120** based on various conditions or information pertaining to vehicle status, vehicle location, and based on driver or operator control or preferences.

(11) The aerodynamic device assembly **120** of example embodiments may be dynamic and intelligently controlled. The “intelligently controlled” nature of the aerodynamic device assembly **120** refers to the fact that automatic adjustments are made to optimize the positioning of the aerodynamic device(s). The “dynamic” nature of the aerodynamic device assembly **120** refers to the fact that it is enabled to be dynamically repositioned responsive to real time conditions and status information to actively improve the performance of the vehicle **110**. Thus, for example, when the operator has enabled automatic control of the aerodynamic device assembly **120** by selecting a mode in which the aerodynamic control system **100** is operable, the state of the aerodynamic device(s) may be controlled based on the current location and conditions experienced by the

vehicle **110**. Thus, for example, various changes in conditions or vehicle status may dynamically cause the state of the aerodynamic device(s) to be changed based on programmed rules or algorithms for control of the aerodynamic device assembly **120**.

(12) In an example embodiment, the aerodynamic control system **100** may include a repositioning assembly (e.g., actuator **130**), which may be provided as part of the aerodynamic control system **100** in order to reposition or move the aerodynamic device assembly **120** under the control of a controller **140** (or control module) of the aerodynamic control system **100**. In some cases, the controller **140** may be part of a vehicle dynamic module (VDM) or other control system of the vehicle **110** that is configured to perform other tasks related or not related to aerodynamic control or performance management. However, the controller **140** could be a dedicated or standalone controller in some cases. The operator (or driver) may disable or enable operation of the actuator **130** by using a user interface **132**. For example, in one or more modes, the actuator **130** may be disabled, and in one or more other modes, the actuator **130** may be enabled to move under control of the controller **140**. As an alternative to having the mode disable or enable the actuator **130** directly, the actuator **130** could be controlled indirectly via the controller **140**. Thus, the user interface **132** may interact with the controller **140** instead of the actuator **130** in some cases. The user interface **132** may be internal or external relative to the vehicle **110**. Thus, for example, when internal, the user interface **132** may include switches, levers, buttons, a mouse, touch screen display, or any other suitable human machine interface (HMI). Also as examples, when external, the user interface **132** may be an external control console, a smart phone, or other interface component that can wirelessly communicate with the vehicle **110**.

(13) In an example embodiment, the controller **140** may receive information that is used to determine vehicle status (or environmental conditions) and vehicle location from or associated with various components or subassemblies **150** of the vehicle **100**. Additionally or alternatively, various sensors that may be operably coupled to the components or subassemblies **150** may be included, and may provide input to the controller **140** that is used in determining vehicle status and location. Such sensors may be part of a sensor network **160** and sensors of the sensor network **160** may be operably coupled to the controller **140** (and/or the components or subassemblies **150**) via a vehicle communication bus (e.g., a controller area network (CAN) bus) **170**.

(14) The components or subassemblies **150** may include, for example, a brake assembly and/or a wheel assembly of the vehicle **110**. Sensors associated with the brake assembly may provide inputs such as brake pedal position, or brake pressure, to the controller **140**. Sensors associated with the wheel assembly may provide information about vehicle speed, wheel angle, etc. Other sensors of the sensor network **160** that may be operably coupled to the brake assembly, the wheel assembly or other parts of the vehicle **110** may provide information relating to brake torque, brake torque rate, vehicle acceleration, individual wheel speeds/angles, etc. Other examples of the components or subassemblies **150** and/or corresponding sensors of the sensor network **160** may provide information relating to pitch, yaw, pitch rate, yaw rate, lateral G force, throttle position, selector button positions associated with chassis and/or vehicle control selections, etc. Thus, for example, the sensors may include an inclinometer, gyroscope, accelerometer, and/or the like. In some cases, the sensor network **160** may further include sensors for detecting weather conditions, road conditions or the like. For example, moisture sensors and temperature sensors may detect wet or icy roads.

(15) Road conditions can also be determined using cameras, microphones, rain sensors, and/or the like in some cases. Thus, for example, the sensor network **160** may also include one or more cameras that operate in the visible light spectrum, infrared cameras, Lidar, and/or the like to detect debris or other obstacles or road conditions. In some cases, however, the cameras may also detect location or position relative to a particular characteristic of the road ahead. For example, the cameras may detect a turn or a straightaway. Moreover, the cameras may detect a magnitude of the turn or length of the straightaway. In an example embodiment, the cameras may also be capable of

comparing image data to previously captured images to determine if the location at which the vehicle **110** currently operates happens to also be a location at which the vehicle **110** had operated in the past. When location correlation occurs, comparisons between prior traversals of the location can be made to determine an optimal setting (or at least a setting to use) for the aerodynamic device assembly **120** at the location (either currently or in the near future). However, location could alternatively be determined using other means. For example, global positioning system (GPS) data or other location sensors could also or alternatively be employed.

(16) Accordingly, for example, the controller **140** may be able to receive numerous different parameters, indications and other information that may be related to or indicative of different situations or conditions associated with vehicle status. The controller **140** may also receive information indicative of vehicle location. The controller **140** may then be configured to use the information received defining vehicle status and/or location in association with the execution of one or more control algorithms that may be used to provide instructions to the actuator **130** in order to control a state (or positions) of the aerodynamic device or devices of the aerodynamic device assembly **120**.

(17) FIG. 2 illustrates a block diagram of various components of the aerodynamic control system **100** of one example embodiment in greater detail. In this regard, for example, the controller **140** may receive status information **200** (e.g., from the sensor network **160**) via the vehicle communication bus **170**. The status information **200** may include any or all of the types of information described above. The controller **140** may also or alternatively receive location information **202** from, for example, either or both of a camera **204** and a GPS device **206**. Processing circuitry (e.g., a processor **210** and memory **220**) at the controller **140** may process the status information **200** and the location information **202** by running one or more control algorithms. The control algorithms may include instructions that can be stored by the memory **220** for retrieval and execution by the processor **210**. In some cases, the memory **220** may further store one or more tables (e.g., look up tables) and various calculations and/or applications may be executed using information in the tables, the status information **200** and/or the location information **202**.

(18) The processor **210** may be configured to execute the control algorithms in series or in parallel. However, in an example embodiment, the processor **210** may be configured to execute multiple control algorithms either based on prior selection or in parallel (e.g., simultaneously). In an example embodiment, each control algorithm (e.g., Alg. 1, Alg. 2, Alg. 3 and Alg. 4 in FIG. 2) may be associated with a different mode, situation, or set of conditions. Alternatively, each of the control algorithms may be associated with a different driving style or strategy that the driver can select based on preference. In any case, a selected one or each of the control algorithms may be executed to determine a corresponding position request (e.g., PR 1, PR 2, PR 3 and PR 4) that informs the actuator **130** where/how to position an aerodynamic device **230** (or devices) of the aerodynamic device assembly **120**. The processor **210** may have a speed and processing power sufficient to ensure that the status information **200** and location information **202** received via the vehicle communication bus **170** are both received and processed in real time (or near real time) to cause corresponding changes to be made to the actuator **130**.

(19) In some cases, to select a mode of operation of the controller **140**, which may influence or dictate which algorithm(s) to use, a mode selector **240** (e.g., a portion or component of user interface **132**) may be provided. The mode selector **240** may have a first mode in which the controller **140** operates automatically as described herein (e.g., a track mode, or high performance mode) and a second mode in which the controller **140** does not operate to adjust positions of the aerodynamic device assembly **120**. However, other modes are also possible.

(20) In an example embodiment, as shown in FIG. 2, the actuator **130** may include a motion source **250** (e.g., an electric motor, a hydraulic actuator, a pneumatic actuator, or the like) that is operably coupled to linkages **252** that are in turn operably coupled to the aerodynamic device **230** (or devices) of the aerodynamic device assembly **120**. Thus, for example, the linkages **252** may include

mechanical structures (e.g., hinges, links, arms, rods, shape-changing materials, etc.) that operably couple motion from an output of the motion source **250** to the aerodynamic device **230** (or devices). In cases where the aerodynamic device **230** is a single rear wing, one or multiple instances of the motion source **250** and linkages **252** may be used to reposition the wing. In other cases in which multiple aerodynamic devices **230** are used (e.g., devices specific to individual wheels or corners of the vehicle **110**), corresponding multiple motion sources **250** and linkages **252** (and perhaps also actuators **130**) may be employed under the control of the controller **140**.

(21) As noted above, the controller **140** may execute one or multiple control algorithms at a time to control the aerodynamic device assembly **120** in a situationally aware manner. In some cases, each of the control algorithms (e.g., Alg. 1, Alg. 2, Alg. 3 and Alg. 4) may have an association with a different situation or scenario. FIG. 3 illustrates a block diagram of one example in which situationally specific control algorithms may be employed. In this regard, as shown in FIG. 3, the controller **140** may receive the location information **202**, the status information **200**, and/or user input **300**. The user input **300** may be made via the selector **240**, or any other suitable input device. The controller **140** may then perform calculations associated with the control algorithms which, in this example, include a low speed corner algorithm **310**, a mid speed corner algorithm **320**, a high speed corner algorithm **330** and a straightaway algorithm **340**.

(22) The status information **200** may, as noted above, be provided from sensors of the sensor network **160**. Inputs such as sound information, moisture data, steering angle, vehicle speed, brake pressure, wheel speed, ride height, etc., from various components including corresponding microphones, rain sensors, accelerometers, pitot tubes, gyroscopes, suspension plots, and/or the like may be included among the status information **200**. The controller **140** may receive the status information **200** and the location information **202** to infer, determine or confirm a segment classification, rating or characterization. In some cases, the use of video data from the camera **204** may be employed to confirm classifications, or to make determinations regarding future positions of the vehicle **110** (e.g., for upcoming corners or turns). The user input **300** may include tuning inputs (e.g., for setting weighting factors involved in each control algorithm), mode selections (e.g., aggressive/non-aggressive, loose/tight, etc. in terms of oversteer/understeer, steering/brake inputs, suspension, etc.) or otherwise enable the operator/driver to have an influence over the operation of the controller **140**. The mode selections may be named for the control paradigm associated with each respective mode, or may be parts of other modes that have other characteristics as well such as the non-limiting examples of Track Mode, Sport Mode, Drift Mode, and/or the like.

(23) In an example embodiment, the location information **202** may be GPS data or other information that enables the controller **140** to determine a specific location of the vehicle **110**. For racing applications, the location information **202** may indicate a track position on a known track, or even a track that is being learned on the fly (e.g., by mapping and recording data for subsequent laps). Either using a known track, or by learning the geometries of the track on which the vehicle **110** is operating, the speed of the vehicle **110** and the magnitude of the various turns encountered at any given time may be used to determine how to characterize each portion of the track. Thus, for example, the controller **140** may monitor movement of the vehicle **110** around the track and determine a characterization, classification or rating for each segment or portion of the track. In one example, the segments may be determined to be either a high speed, low speed, or mid speed turn or corner, or a straightaway. However, other classifications of segments are also possible, and control algorithms may exist for each respective one of the classifications that are possible.

(24) If a particular segment is characterized as a low speed corner, then the low speed corner algorithm **310** may be used by the controller **140** to determine how to control the aerodynamic device assembly **120** in the corresponding segment. If a particular segment is instead characterized as a mid speed corner, then the mid speed corner algorithm **330** may be used by the controller **140** to determine how to control the aerodynamic device assembly **120** in the corresponding segment. Likewise, if a particular segment is characterized or classified as either a high speed corner or a

straightaway, then the high speed corner algorithm **330** or the straightaway algorithm **340**, respectively, may be used by the controller **140** to determine how to control the aerodynamic device assembly **120** in the respective segment.

(25) As shown in FIG. **3**, the low speed corner algorithm **310** may favor a forward aero balance and maximum downforce strategy **312**, which may be programmed as part of the low speed corner algorithm **310**. This may translate into a command sequence or position request (i.e., PR1) to the actuator **130** for changing the aerodynamic device **230** of the aerodynamic device assembly **120** to achieve the balance desired to provide better turn in. This (and other algorithms) may, in some cases, operate by defining target parameters or positions for the aerodynamic device **230** and measuring error from such parameters/positions with continued effort to drive the error to a minimum value. Similarly, the mid speed corner algorithm **320** may favor a mid aero balance and maximum downforce strategy **322**, which may be programmed as part of the mid speed corner algorithm **320**. This may translate into a command sequence or position request (i.e., PR2) to the actuator **130** for changing the aerodynamic device **230** of the aerodynamic device assembly **120** to achieve the balance desired. For the high speed corner algorithm **330**, a rear aero balance maximum downforce strategy **332** may be employed, which may translate into a command sequence or position request (i.e., PR3) to the actuator **130** for changing the aerodynamic device **230** of the aerodynamic device assembly **120** to achieve the balance desired and improved stability. Finally, for the straightaway algorithm **340**, a minimum drag, no balance strategy **342** may be employed, which may translate into a command sequence or position request (i.e., PR4) to the actuator **130** for changing the aerodynamic device **230** of the aerodynamic device assembly **120** to achieve minimum drag.

(26) As can be appreciated from the descriptions above, location information **202** obtained from cameras and/or other location sensors may be key in identifying location for either learning new locations that will be repeated, or confirming location on a known track or geographical area (e.g., based on a map). The location information **202** can therefore be used to classify current or future driving segments. The classification can sometimes be entirely geography-based (i.e., based on map location and correspondence to known locations that are already classified). However, in other cases, the status information **200** may be used to facilitate classifying a segment based on the speeds and accelerations that are experienced for the corresponding location. The location information may then just confirm revisiting the same location that was otherwise classified based on actually experienced parameters at the location. Thus, segment classification can be either experiential (based on measured parameters) or location based.

(27) In an example embodiment, as noted above, the operator/driver may provide user input **300** to modify the algorithms, or provide information that may be used to weight various variables that may be used in each algorithm. In this regard, for example, the control algorithms described above may each have baseline settings that may correspond to the strategies and balance considerations described above. However, the operator/driver may modify the strategies or replace strategies with other strategies according to the preferences of the operator/driver. In some cases, the memory **220** may store a plurality of algorithms with corresponding baseline strategies. The memory **220** may also store different strategies that the operator/driver may parse through to select strategies of interest and pair such strategies with respective control algorithms for each segment classification. By this method, the operator/driver may tailor the vehicle response to his/her own preferences by dictating what strategies are to be used in each segment rather than using the baseline strategies that are otherwise predefined for each segment. Instead of wholesale changes to strategies, the operator/driver may also or alternatively modify curves, offsets, values, parameters, and/or the like within strategies to fine tune the strategies.

(28) Each algorithm may set different target positions or values for the aerodynamic device **230** based on a set of calculations or determinations intended to optimize parameters associated with the situation in which the vehicle **110** is or will find itself soon. In some cases, the algorithms may

include tables of values or positions for the aerodynamic device **230**, where the tables are entered using specific vehicle parameters (e.g., speed, brake pressure, etc.). Various other vehicle parameters may be entry values into such tables or may be weighting factors, or modifiers to either the table entry values or the resulting values or positions of the aerodynamic device **230**. For example, ride height, yaw rate, tire age, vehicle weight, and/or the like may be modifiers in the manner described above. Calculations may also consider various vehicle status conditions (e.g., weight, weight distribution, passengers, loading, fuel status, tire condition, etc.) and weather conditions (e.g., ambient external vehicle temperature, wet roads, icy roads, etc.) to tailor handling to the conditions of the road surface at any given time.

(29) In an example embodiment, the control algorithms may include iterative strategies for achieving target setting in real time. However, some examples embodiments may further employ the ability to improve the algorithm performance over time as well. In this regard, for example, the controller **140** may employ machine learning to adaptively improve, particularly in situations where prior data is available for the same location or situation.

(30) FIG. **4** illustrate a control flow diagram for adaptive algorithm improvement in accordance with an example embodiment. In this regard, the controller **140** may determine whether a current location of the vehicle corresponds to a known location at operation **400**. To accomplish such determination, the controller **140** may compare the current location of the vehicle to stored prior locations, or a database of known locations (e.g., due to preloaded maps, racetrack information, or the like). If the current location is known, the controller **140** may then load path segment information for the current location at operation **410**. In some cases, operation **410** may further include sequentially loading additional path segment information for a series of path segments that extend from the current location as well. The path segments may be loaded either as reached, or immediately prior to reaching each respective segment. This effectively gives the controller **140** an understanding of the best strategy to employ for control of aerodynamic devices for the current segment (and future segments), and a control algorithm for the current path segment may then be selected at operation **420**. The aerodynamic device assembly **120** may then be controlled by the controller **140** based on the selected control algorithm at operation **430**. In some cases, a loop may be formed including confirming a next segment at operation **440** prior to returning to operation **410**, or simply returning to operation **400**.

(31) As noted above, the control of the aerodynamic device assembly **120** may include table entry and modifications that are based on the strategy selected, and the real time driving parameters encountered during traversal of the path segment. In some embodiments, the controller **140** may have a target understeer or oversteer value for a turn, and the control algorithm may operate to facilitate reaching the target by reducing error between current understeer or oversteer values and the target values. The aggressiveness with which the error between target and actual values is reduced may be one of the factors that differentiates respective different control algorithms. In other words, the error between target and actual values may have different closure rates in which higher rates of closure correspond to more aggressive error reduction and lower rates of closure correspond to less aggressive error reduction.

(32) Meanwhile, during performance of the operation **430**, all data associated with traversal of the path segment may be recorded in association with the path segment at operation **450**. Thereafter, data associated with the same or similar segments may be compared and analyzed to determine whether to update either the control algorithms themselves (e.g., fine tuning of algorithms) or selection criteria used by the controller **140** to select control algorithms for a given path segment at operation **460**. The updating may be automatic (e.g., using machine learning), or may be aided by input from the driver/operator. In this regard, for example, the selector **230** may be part of a user interface that enables the driver/operator to make adjustments to control algorithms or parameters/variables associated therewith as described above.

(33) If instead the current location was not known as a result of operation **400**, then all data

associated with traversal of the path segment may be recorded in association with the path segment at operation **470**. Operation **470** effectively takes a previously unknown path segment and makes it a known path segment. Data regarding the vehicle's performance over the path segment and data regarding the path segment itself may then be analyzed to classify the path segment (e.g., as a high speed, low speed, or mid speed corner, as a straightaway, or any other suitable path segment classifications) at operation **480**. After operation **480**, the corresponding data for the path segment is available for comparison and usage the next time the path segment is encountered (or a similar path segment is encountered).

(34) In some embodiments, path segments may further include sub-segments that may have respective different strategies (or sub-strategies) associated therewith for each respective control algorithm. FIG. 5 illustrates one example. In this regard, FIG. 5 illustrates a section of roadway (or track) that includes a significant turn. The vehicle may approach the turn on a straightaway path segment **500** and the straightaway algorithm **340** may be employed in this path segment. However, either via GPS, or via camera usage, the vehicle may determine that the corner is being approached, and a control algorithm change (and corresponding control strategy change) is appropriate. In this case, a physical feature **510** (e.g., a tree or other object that is distinct) may be detected by the camera to help determine location on the track or roadway, and that a corner associated with the turn mentioned above is being approached.

(35) The turn may be of sufficient magnitude to qualify the turn as a low speed turn. Thus, upon approaching the turn, and recognizing the same, the low speed corner algorithm **310** may be selected. However, as noted above, the low speed corner algorithm **310** may include sub-strategies or algorithms that are employed based on where the vehicle is within the changing phases of making the turn. In this regard, for example, a braking zone **520** may exist at the end of the straightaway path segment **500** to allow the vehicle time to slow down before reaching turn-in zone **530**, where the handwheel is turned to begin turning the vehicle. In the braking zone **520**, a braking zone strategy **522** may be employed to position the aerodynamic device **230** to provide forward balance to add vertical loading and increase front tire performance envelope. Over time, positions of the aerodynamic device **230**, and corresponding performance parameters may be recorded and modified slightly to (again via machine learning) optimize performance in this (and similar) sub-segments.

(36) When the turn-in zone **530** is reached, a turn-in strategy **532** may be employed to continue to position the aerodynamic device **230** to provide forward balance to increase front tire performance envelope and provide better turn-in response. Thereafter, in the mid corner zone **540**, a mid corner strategy **542** may be employed to position the aerodynamic device **230** to balance stability and maximum lateral acceleration. Finally, when corner exit zone **550** is reached, a corner exit strategy **552** may be employed to position the aerodynamic device **230** to provide rearward aerodynamic balance as the rear tire load is increased due to longitudinal acceleration before exiting the turn entirely and returning to the straightaway algorithm **340** in straightaway path segment **560**.

(37) Based on the descriptions above, it should be appreciated that positioning of an aerodynamic device may be intelligently managed based on applying different selectable control algorithms for different classifications of road segment. The road segment the vehicle is on or approaching may therefore be determined (e.g., via GPS, cameras, and/or the like) to either classify the segment (if it is being traversed for a first time) or recognize an already classified segment. Once classified, selection may be enabled for a strategy-specific control algorithm for traversing each type of segment. The selected control algorithm positions the aerodynamic device, and various parameters are recorded that effectively measure vehicle performance over the segment. This stored data (which is location specific or segment type specific) can then be analyzed to improve performance (e.g., efficiency, lap time, etc.) to try to achieve optimal aerodynamic efficiency and/or balance for a given segment (or similar segment). Moreover, machine learning may be applied to compare lap times or performance at the same location across different traversals to determine potential better

balance or efficiency results that may be achieved with different aerodynamic device positions. Over multiple traversals, fine tuning may be accomplished.

(38) Example embodiments may use input data from vehicle sensors, which may include status information such as sound information, moisture data, ride height, etc. or vehicle performance data such as steering angle, vehicle speed, brake pressure, wheel speed, accelerometer data, etc. The input data may be processed, and a machine learning algorithm may be employed to determine ideal or at least improved aerodynamic efficiency and balance to optimize lap time and drivability on track, and maximize performance on-road. As noted above, the outputs of example embodiments may be tuned for specific corners or segments of known locations or tracks. Outputs may also be optimized to reduce drag in non-grip limited conditions, or may be adjusted for driver preference (tight car, loose car, etc.). Example embodiments may use outputs to optimize handling for varying track and weather conditions, and for varying vehicle conditions (e.g., weight and weight distribution, full and empty fuel, tire state, passenger weight or loading, etc.).

(39) Accordingly, an aerodynamics control system for a vehicle may be provided. The aerodynamics control system may include a repositionable aerodynamic device disposed at a portion of the vehicle, a controller operably coupled to components and/or sensors of the vehicle to receive information including vehicle performance data and position information for the aerodynamic device, and a vehicle location sensor determining location information for the vehicle. The controller stores the vehicle performance data and the position information in association with the location information for each of a plurality of locations. Responsive to detecting an approach of the vehicle to one of the locations, the controller provides a control instruction to position the aerodynamic device based on recorded vehicle performance data and recorded position information associated with the one of the locations.

(40) The system of some embodiments may include additional features, modifications, augmentations and/or the like to achieve further objectives or enhance performance of the system. The additional features, modifications, augmentations and/or the like may be added in any combination with each other. Below is a list of various additional features, modifications, and augmentations that can each be added individually or in any combination with each other. For example, the location may be a path segment, and the controller may determine the control instruction based on a selected one of a plurality of control algorithms executable by the controller. The selected one of the plurality of control algorithms may be selected based on a classification of the path segment. In an example embodiment, each of the plurality of control algorithms may include instructions defining a strategy for positioning the aerodynamic device for a particular type of path segment with which respective ones of the plurality of control algorithms are associated. In some cases, the types of path segment may include a straightaway and one or more corner types classified by a speed at which a corner is traversed by the vehicle. In an example embodiment, at least one of the types of path segment includes sub-segments defining phases for entering into, traversing and exiting out of the corner, and the instructions include a balance strategy for positioning the aerodynamic device in each of the phases. In some cases, the controller may employ machine learning to modify the control algorithms based on the vehicle performance data and the position information for the aerodynamic device after each subsequent traversal of the one of the locations. In an example embodiment, the system includes a camera to provide visual information to classify the path segment. In some cases, the classification of the path segment may be performed based on a comparison of the location information to previously traversed locations for which the recorded vehicle performance data and the recorded position information are stored. In an example embodiment, the classification of the path segment may be performed based on experiential data from a prior traversal of the path segment. In some cases, the classification of the path segment may be performed based on a combination of the location information and experiential data from a prior traversal of the path segment. In an example embodiment, the control algorithm may be modifiable based on user input to define vehicle performance preferences. In

some cases, selection criteria for selecting the control algorithm may be modifiable based on user preferences. In an example embodiment, the control algorithm may be modifiable based on vehicle weight, weight distribution or weather conditions.

(41) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. An aerodynamics control system for a vehicle, the system comprising: a repositionable aerodynamic device disposed at a portion of the vehicle; a controller operably coupled to components or sensors of the vehicle to receive information including vehicle performance data and position information for the aerodynamic device; and a vehicle location sensor determining location information for the vehicle, wherein the controller stores the vehicle performance data and the position information in association with the location information for each of a plurality of locations, wherein, responsive to detecting an approach of the vehicle to one of the locations, the controller provides a control instruction automatically that dynamically repositions the aerodynamic device based on recorded vehicle performance data and recorded position information associated with the one of the locations, wherein the controller determines and executes the control instruction based on a selected one of a plurality of control algorithms run by the controller simultaneously, wherein the selected one of the plurality of control algorithms is selected as an optimal one of the plurality of control algorithms being run simultaneously to deliver an optimal control instruction based on a classification of a path segment, and wherein the classification of the path segment is performed in real-time using present vehicle status information, wherein each of the plurality of control algorithms includes instructions to control positioning the aerodynamic device for a particular type of path segment with which respective ones of the plurality of control algorithms are associated, and wherein the types of path segment include a straightaway and a plurality of corner types classified by a speed at which a corner is traversed by the vehicle.
2. The system of claim 1, wherein at least one of the types of path segment further comprises sub-segments defining phases for entering into, traversing and exiting out of the corner, and wherein the instructions comprise a balance strategy for positioning the aerodynamic device in each of the phases.
3. The system of claim 1, wherein the controller employs machine learning to modify the control algorithms based on the vehicle performance data and the position information for the aerodynamic device after each subsequent traversal of the one of the locations.
4. The system of claim 1, wherein the system comprises a camera to provide visual information to

classify the path segment.

5. The system of claim 1, wherein the classification of the path segment is performed based on a comparison of the location information to previously traversed locations for which the recorded vehicle performance data and the recorded position information are stored.

6. The system of claim 1, wherein the classification of the path segment is performed based on experiential data from a prior traversal of the path segment.

7. The system of claim 1, wherein the classification of the path segment is performed based on a combination of the location information and experiential data from a prior traversal of the path segment.

8. The system of claim 1, wherein the control algorithm is modifiable based on user input to define vehicle performance preferences.

9. The system of claim 1, wherein selection criteria for selecting the control algorithm is modifiable based on user preferences.

10. The system of claim 1, wherein the control algorithm is modifiable based on vehicle weight, weight distribution or weather conditions.

11. A method of providing vehicle stability control to a vehicle having an aerodynamic device disposed on the vehicle, the method comprising: receiving status information and location information from components or a sensor network of the vehicle; determining whether the location information is a known location; recording the location information and status information as path data for a path segment corresponding to the location information, and classifying the path segment in response to the location information not being known; and controlling the aerodynamic device automatically based on a selected control algorithm in response to the location information being known, wherein the selected control algorithm is selected based on a classification of the path segment, wherein the selected control algorithm defines positions of the aerodynamic device and measures error based on the positions of the aerodynamic device to drive the error to a minimum value, and wherein the classification of the path segment is performed in real-time exclusively using present vehicle status information obtained at the vehicle in real-time, wherein each of the plurality of control algorithms includes instructions to control positioning the aerodynamic device for a particular type of path segment with which respective ones of the plurality of control algorithms are associated, and wherein the types of path segment include a straightaway and a plurality of corner types classified by a speed at which a corner is traversed by the vehicle.

12. The method of claim 11, wherein classifying the path segment is performed based on the location information or experiential data from a prior traversal of the path segment.

13. The method of claim 11, wherein classifying the path segment is performed based on a combination of the location information and experiential data from a prior traversal of the path segment.

14. The method of claim 11, wherein determining whether the location information is known comprises comparing the location information to previously traversed locations for which the status information was recorded.

15. The method of claim 11, wherein the selected control algorithm is modifiable based on user input to define vehicle performance preferences.

16. The method of claim 11, wherein selection criteria for selecting the selected control algorithm are modifiable based on user preferences.

17. The method of claim 11, wherein the control algorithm is modifiable based on vehicle weight, weight distribution or weather conditions.

18. An aerodynamics control system for a vehicle, the system comprising: a repositionable aerodynamic device disposed at a portion of the vehicle; a controller operably coupled to components or sensors of the vehicle to receive information including vehicle performance data and position information for the aerodynamic device; and a vehicle location sensor determining location information for the vehicle, wherein the controller stores the vehicle performance data and

the position information in association with the location information for each of a plurality of locations, wherein, responsive to detecting an approach of the vehicle to one of the locations, the controller provides a control instruction automatically that dynamically repositions the aerodynamic device based on recorded vehicle performance data and recorded position information associated with the one of the locations, wherein the controller determines the control instruction based on a selected one of a plurality of control algorithms executed by the controller, wherein the selected one of the plurality of control algorithms is based on a classification of a path segment, and wherein each of the plurality of control algorithms includes instructions to control positioning of the aerodynamic device for a particular type of the path segment with which respective ones of the plurality of control algorithms are associated, wherein the positioning of the aerodynamic device is controlled to adjust aerodynamic balance, downforce, or drag based on the classification of the path segment, wherein the positioning of the aerodynamic device is controlled to increase downforce based on the classification of the path segment being a corner traversal, and wherein the classification of the path segment is performed in real-time exclusively using present vehicle status information obtained at the vehicle in real-time.
